

Fig. 31.1. Kids playing baseball inside the space colony, as seen by Cooper looking through a window. *[From* Interstellar*, used courtesy of Warner Bros. Entertainment Inc.]*

How did it get lifted into space? The key, of course, was the quantum data (in my scientist's interpretation, the *quantum gravity laws*) that TARS extracted from Gargantua's singularity (Chapters 26 and 28) and Cooper transmitted to Murph (Chapter 30).

In my interpretation, by discarding quantum fluctuations from those laws (Chapter 26), Murph learned the nonquantum laws that govern gravitational anomalies. And from those laws, she figured out how to control the anomalies.

As a physicist, I'm eager to know the details. Was Professor Brand on the right track in the equations that covered his blackboards? (Chapter 25 and this book's page at Interstellar.withgoogle.com.) Did he really have half the answer, as Murph asserted before getting the